

SOP-CAL-2800: Instrumentation Calibration

Document owner: Instrumentation Department **Effective date:** 2024-02-05 **Review cycle:** 24 months

Section 1. Purpose and Scope

Governs calibration frequency, tolerance, and out-of-tolerance handling for process instrumentation, with emphasis on safety-instrumented and custody-transfer devices.

Section 2. Calibration Frequency

Clause 2.1 — General process instrumentation is calibrated every 12 months; safety-instrumented system (SIS) devices follow the interval specified in the site's SIL verification calculation, which may be shorter.

Clause 2.2 — Custody-transfer metering instruments are calibrated per the applicable regulatory/contractual schedule, independent of the general 12-month cycle.

Section 3. Tolerance and As-Found/As-Left

Clause 3.1 — Both as-found and as-left readings are recorded at every calibration, at minimum 3 points spanning the instrument's operating range.

Clause 3.3 — An as-found reading outside the device's acceptance tolerance is a calibration failure requiring investigation into the cause (drift, damage, environmental factor) before the device is returned to service, not simply re-adjusted and closed out.

Section 4. Safety-Instrumented Devices

Clause 4.1 — A SIS device found out of tolerance at calibration is treated as a potential demand-on-safety-function event; the Process Safety Department is notified and the associated safety function's availability during the out-of-tolerance period is evaluated.

Section 5. Records

Clause 5.1 — Calibration certificates are filed to the Instrumentation File and retained for the life of the device; SIS device calibration records are retained per the SIL verification document's specified retention.